

■ COMPLETE NOTES - TOPIC 08

CSS Shadows and Outlines

Box shadow, text shadow, drop shadow aur outlines - Basic se Advanced

Practical Level tak

Web Development Course - Shauray

box-shadow

text-shadow

filter: drop-shadow

inset shadow

multiple shadows

outline

outline-offset

focus-visible

border vs outline

performance

CSS Shadows Kya Hote Hain?

Depth, elevation, glow aur visual separation create karne ka system

SIMPLE DEFINITION

CSS shadow ek visual effect hai jo element ke around ya text ke behind depth create karta hai.

Shadow se UI mein card lifted feel, button focus, glow effect, neumorphism, depth hierarchy aur readability improve ho sakti hai.

Shadows ka main purpose decoration nahi, visual communication hai. Agar card background se separate nahi dikh raha, shadow help kar sakti hai. Agar text image ke upar readable nahi hai, text-shadow help kar sakta hai. Agar icon transparent PNG/SVG shape ke around shadow chahiye, drop-shadow useful hota hai.

■ CSS Mein Shadow Ke Main Types

Type	Property	Use
Box Shadow	box-shadow	Cards, modals, buttons, panels, containers
Text Shadow	text-shadow	Headings, hero text, glow text, text readability
Drop Shadow	filter: drop-shadow()	Transparent images, icons, SVG shapes
Inner Shadow	box-shadow: inset	Pressed buttons, input depth, inner glow

SHADOW VISUAL TYPES

soft
box-shadow

inset
shadow

glow
shadow

■ **Golden Start:** Shadow tab best lagti hai jab woh subtle ho. Bahut dark, hard ya oversized shadow UI ko cheap feel kara sakti hai.

box-shadow Property

Element ke box ke around shadow add karna

DEFINITION

box-shadow element ke border box ke around shadow create karta hai. Isme horizontal offset, vertical offset, blur radius, spread radius, color aur optional **inset** value use hoti hai.

box-shadow syntax

```
.card {  
  box-shadow: offset-x offset-y blur spread color;  
}  
  
.card {  
  box-shadow: 0 12px 30px 0 rgba(0, 0, 0, 0.25);  
}
```

■ box-shadow Values Explained

Value	Meaning	Example
offset-x	Shadow left/right move hoti hai. Positive right, negative left.	10px, -10px
offset-y	Shadow up/down move hoti hai. Positive down, negative up.	12px, -6px
blur-radius	Shadow kitni soft/blurred hogi. 0 means sharp.	20px
spread-radius	Shadow size expand ya shrink karta hai. Optional hota hai.	4px, -2px
color	Shadow ka color. Usually rgba/hsla best hota hai.	rgba(0,0,0,.25)
inset	Shadow element ke andar dikhti hai.	inset 0 2px 8px #000

■ Offset Direction Example

● ● ● shadow direction

```
.right-down {  
  box-shadow: 10px 10px 20px rgba(0, 0, 0, 0.3);  
}  
  
.left-up {  
  box-shadow: -10px -10px 20px rgba(0, 0, 0, 0.3);  
}
```

■ **Practical Tip:** Realistic shadows usually bottom side par zyada hoti hain: `0 8px 24px`
`rgba(0,0,0,.18)`. Kyunki light generally top se imagine hoti hai.

Soft, Hard, Spread and Inset Shadows

Shadow ke different visual styles aur unka use

1. Soft Shadow

Soft shadow mein blur zyada hota hai aur opacity low hoti hai. Modern cards, modals aur dropdowns ke liye yeh most common hai.

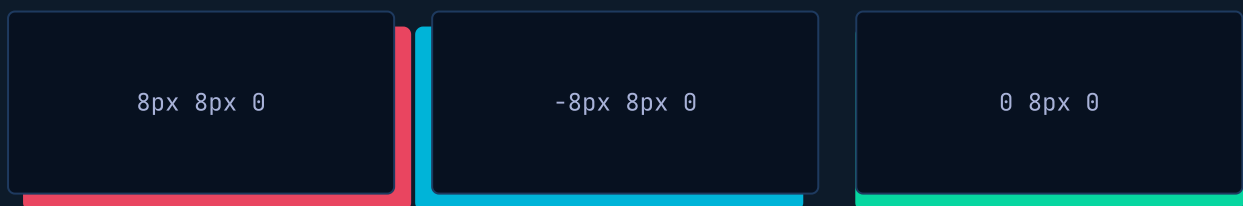
● ● ● soft shadow

```
.card {  
  box-shadow: 0 16px 40px rgba(0, 0, 0, 0.18);  
}
```

2. Hard Shadow

Hard shadow mein blur o hota hai. Yeh editorial, playful, retro ya brutalist design mein useful hoti hai.

HARD SHADOW DEMO



3. Spread Shadow

Spread radius shadow ko expand ya shrink karta hai. Positive spread shadow ko bada karta hai, negative spread shadow ko tight/controlled banata hai.

spread shadow

```
.wide-shadow {  
  box-shadow: 0 10px 30px 8px rgba(0, 0, 0, 0.22);  
}  
  
.tight-shadow {  
  box-shadow: 0 16px 30px -12px rgba(0, 0, 0, 0.45);  
}
```

▪ 4. Inset Shadow

inset shadow element ke andar render hoti hai. Yeh pressed button, input field inner depth, dark panel inner glow ya engraved effect ke liye useful hai.

inset shadow

```
.input {  
  box-shadow: inset 0 2px 8px rgba(0, 0, 0, 0.25);  
}
```

Multiple Box Shadows

Comma separated shadows se realistic depth aur layered effects banana

DEFINITION

Ek element par multiple shadows laga sakte hain. Har shadow comma se separate hoti hai. Pehli shadow top layer par paint hoti hai, baaki shadows uske neeche layer hoti hain.

multiple shadows

```
.premium-card {  
  box-shadow:  
    0 2px 8px rgba(0, 0, 0, 0.12),  
    0 16px 40px rgba(0, 0, 0, 0.20);  
}
```

■ Why Multiple Shadows?

Single shadow sometimes flat lagti hai. Real world mein object ke paas contact shadow aur ambient shadow dono hoti hain. Contact shadow chhoti/dark hoti hai, ambient shadow badi/soft hoti hai. Dono combine karke realistic elevation milta hai.

Layer	Style	Purpose
Contact shadow	Small blur, darker opacity	Object surface ke close feel hota hai
Ambient shadow	Large blur, low opacity	Natural soft depth create hoti hai
Highlight shadow	Light color or inset	Raised/glass/neumorphic effect

■ Material Style Elevation Example


```
.elevation-1 {  
  box-shadow: 0 1px 3px rgba(0,0,0,.12), 0 1px 2px rgba(0,0,0,.16);  
}  
  
.elevation-3 {  
  box-shadow: 0 10px 20px rgba(0,0,0,.18), 0 6px 6px rgba(0,0,0,.14);  
}
```

■ **Pro Tip:** Shadow scale design system ka part banao: low, medium, high, modal. Har jagah random shadows use karne se UI inconsistent lagti hai.

text-shadow Property

Text ke behind shadow, glow ya readability layer add karna

DEFINITION

text-shadow text glyphs ke behind shadow create karta hai. Isme offset-x, offset-y, blur-radius aur color use hota hai. Isme spread radius nahi hota.

TEXT SHADOW DEMO

CSS Text Shadow

text-shadow syntax

```
h1 {  
  text-shadow: offset-x offset-y blur color;  
}  
  
.hero-title {  
  text-shadow: 0 3px 14px rgba(0, 0, 0, 0.55);  
}
```

■ Text Shadow Use Cases

Use Case	Shadow Style	Reason
Text on image	0 2px 8px rgba(0,0,0,.65)	Readability improve hoti hai
Glow text	0 0 18px color	Neon/highlight effect
Emboss effect	1px 1px 0 dark, -1px -1px 0 light	Raised text illusion
Retro title	3px 3px 0 solid color	Hard graphic style

■ Multiple Text Shadows

● ● ● neon text

```
.neon {  
  color: #fff;  
  text-shadow:  
    0 0 6px #00b4d8,  
    0 0 18px #00b4d8,  
    0 0 36px #00b4d8;  
}
```

■ **Readability Warning:** Long paragraphs par heavy text-shadow avoid karo. Text muddy aur tiring ho sakta hai. Text-shadow mostly headings ya image overlays par use karo.

filter: drop-shadow()

Transparent image ya shape ke real visible pixels par shadow lagana

DEFINITION

`filter: drop-shadow()` element ke actual visible shape ke around shadow banata hai. Yeh especially PNG, SVG icons, transparent logos aur irregular shapes ke liye useful hai.

■ box-shadow vs drop-shadow

Feature	box-shadow	drop-shadow()
Shadow shape	Element ke rectangular box ke around	Visible pixels/alpha shape ke around
Spread radius	Supported	Not supported
Multiple shadows	Comma separated	Multiple filter functions chain kar sakte hain
Best for	Cards, boxes, modals	PNG, SVG, icons, cutout images

● ● ● drop-shadow example

```
.logo {  
  filter: drop-shadow(0 8px 14px rgba(0, 0, 0, 0.35));  
}  
  
.glowing-icon {  
  filter: drop-shadow(0 0 10px #00b4d8);  
}
```

■ Multiple drop-shadow Filters

```
.icon {  
  filter:  
    drop-shadow(0 2px 4px rgba(0, 0, 0, 0.35))  
    drop-shadow(0 0 14px rgba(0, 180, 216, 0.45));  
}
```

■ **Practical Rule:** Box hai to `box-shadow`. Transparent shape hai to `drop-shadow()`. Text hai to `text-shadow`.

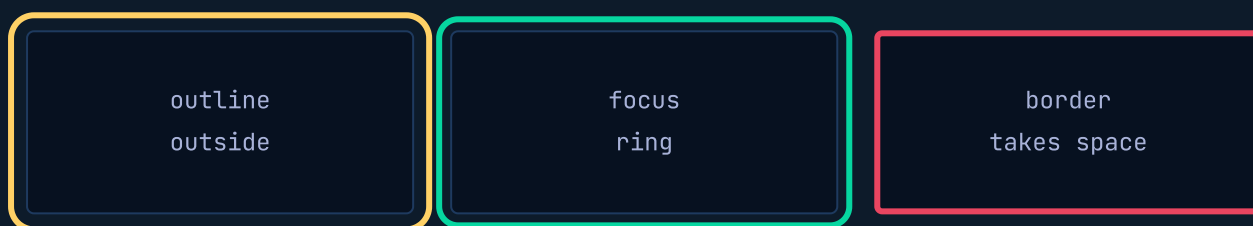
CSS Outline Property

Element ke outside ek non-layout line draw karna

SIMPLE DEFINITION

outline border ke similar ek line hoti hai, but yeh layout space occupy nahi karti. Outline element ke outside draw hoti hai aur focus indicators ke liye very important hai.

OUTLINE VISUAL DEMO



● ● ● outline syntax

```
.button {  
  outline: 3px solid #06d6a0;  
  outline-offset: 4px;  
}
```

■ Outline Longhand Properties

Property	Meaning	Example
<code>outline-width</code>	Outline ki thickness	3px
<code>outline-style</code>	Line style	solid, dashed, dotted
<code>outline-color</code>	Outline ka color	#06d6a0
<code>outline-offset</code>	Outline aur element ke beech distance	4px
<code>outline</code>	Width + style + color shorthand	2px solid red

■ Border vs Outline

Feature	Border	Outline
Layout space	Space occupy karta hai	Space occupy nahi karta
Box model	Width/height calculation affect kar sakta hai	Box model affect nahi karta
Individual sides	border-top/right/bottom/left possible	Individual side outline nahi hota
Border radius	border-radius follow karta hai	Modern browsers mein generally follows, but not same control
Best use	Permanent visual boundary	Focus ring, temporary highlight

Focus Outlines and Accessibility

Keyboard users ke liye visible focus indicator banana

IMPORTANT CONCEPT

Focus outline accessibility ka core part hai. Jab user keyboard se Tab key use karke button, link, input ya control par move karta hai, focus indicator batata hai ki currently active element kaunsa hai.

▪ Default Outline Remove Kyun Dangerous Hai?

Kai beginners `outline: none` laga dete hain because blue outline unko ugly lagti hai. Lekin agar replacement focus style nahi diya, keyboard users ko page use karna mushkil ho jata hai. Isliye outline remove karna allowed hai only if better custom focus style diya gaya ho.

● ● ● bad vs good focus

```
/* Bad: focus invisible ho jayega */
button:focus {
  outline: none;
}

/* Good: custom visible focus ring */
button:focus-visible {
  outline: 3px solid #06d6a0;
  outline-offset: 3px;
}
```

▪ :focus vs :focus-visible

Selector	Meaning	Use
<code>:focus</code>	Element focused hai, mouse ya keyboard kisi bhi reason se	General focus state
<code>:focus-visible</code>	Browser decide karta hai ki focus indicator visible hona chahiye	Keyboard-friendly modern focus rings
<code>:focus-within</code>	Element ya uska child focused hai	Form groups, dropdown containers

■ Accessible Focus System

● ● ● focus ring system

```
:root {
  --focus-ring: #06d6a0;
}

a:focus-visible,
button:focus-visible,
input:focus-visible {
  outline: 3px solid var(--focus-ring);
  outline-offset: 3px;
}
```

■ **Must Remember:** `outline: none` bina replacement ke accessibility bug hai. Professional websites always visible keyboard focus provide karti hain.

Professional Shadow System

Random shadow nahi, consistent elevation scale use karna

Shadow Tokens

Large project mein har component random shadow use kare, to UI inconsistent lagta hai. Better approach hai CSS variables se shadow tokens banana: small, medium, large, focus, glow.

shadow tokens

```
:root {  
  --shadow-sm: 0 2px 8px rgba(0, 0, 0, 0.12);  
  --shadow-md: 0 10px 24px rgba(0, 0, 0, 0.18);  
  --shadow-lg: 0 20px 50px rgba(0, 0, 0, 0.25);  
  --shadow-focus: 0 0 0 4px rgba(6, 214, 160, 0.28);  
}  
  
.card {  
  box-shadow: var(--shadow-md);  
}
```

Elevation Scale

Level	Use	Shadow Feeling
0	Flat sections, base cards	No shadow or border only
1	Subtle cards, inputs	Very soft separation
2	Dropdowns, hover cards	Clearly lifted
3	Modals, popovers	Highest layer, strongest depth

Dark Mode Shadows

Dark UI mein black shadow kam visible hoti hai. Isliye dark mode mein border, subtle glow, lighter surface color aur colored shadow ka combination better work karta hai.

● ● ● dark ui shadow

```
.dark-card {  
  background: #0d1b2a;  
  border: 1px solid rgba(255, 255, 255, 0.08);  
  box-shadow: 0 18px 40px rgba(0, 0, 0, 0.35);  
}
```

■ **Design Rule:** Shadow ko hierarchy ke liye use karo. Jo element user ke closer/above feel hona chahiye, uski shadow stronger ho sakti hai.

Performance, Animation and Common Mistakes

Professional level par shadows ko safely use karna

■ Shadow Performance

Large blur shadows browser ko paint work zyada karwa sakti hain, especially jab many elements par heavy shadow ho ya animation chal rahi ho. Static cards par shadow usually fine hai, but huge animated shadows costly ho sakti hain.

Pattern	Performance	Advice
Small static shadow	Usually fine	Cards/buttons ke liye safe
Very large blur shadow	Paint cost high ho sakta hai	Limit use, test on low-end devices
Animating box-shadow	Expensive ho sakta hai	Opacity/transform based tricks better
Many glowing elements	Costly and noisy	Use sparingly

■ Hover Shadow Best Practice

Hover par shadow animate karne ke saath transform use karna common hai. Smooth effect ke liye `transform` animation best hoti hai, shadow ko small change tak limit rakho.

● ● ● hover elevation

```
.card {  
  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.14);  
  transition: transform 200ms ease, box-shadow 200ms ease;  
}  
  
.card:hover {  
  transform: translateY(-4px);  
  box-shadow: 0 16px 34px rgba(0, 0, 0, 0.22);  
}
```

■ Common Mistakes

MISTAKE

Har card par bahut dark shadow.
Shadow color pure black with high opacity.
Text-shadow paragraphs par lagana.
Focus outline remove kar dena.
box-shadow ko PNG shape shadow ke liye use karna.
Huge glow effects everywhere.

FIX

Subtle rgba shadow use karo.
Elevation scale maintain karo.
Text-shadow mainly headings/image overlays par rakho.
focus-visible custom outline do.
Transparent shape ke liye drop-shadow use karo.
Glow sirf important highlight ke liye use karo.

■ **Final Warning:** Shadows and outlines visual polish ke tools hain, but overuse karne se UI noisy ho jata hai. Subtle, consistent aur accessible approach best hai.

CSS Shadows and Outlines - Complete Summary

Revision ke liye all important concepts ek jagah

Property / Value	Meaning	Best Use
<code>box-shadow</code>	Element box ke around shadow	Cards, modals, dropdowns
<code>text-shadow</code>	Text ke behind shadow	Headings, image overlay text, glow text
<code>filter: drop-shadow()</code>	Visible pixels/shape ke around shadow	Transparent PNG, SVG, icons
<code>inset</code>	Shadow element ke andar	Pressed effect, input inner depth
<code>outline</code>	Element outside non-layout line	Focus rings, temporary highlight
<code>outline-offset</code>	Outline aur element ke beech gap	Clean accessible focus style
<code>:focus-visible</code>	Keyboard-friendly focus selector	Accessible custom outlines
<code>multiple shadows</code>	Comma separated layered shadows	Realistic elevation, glow effects

■ Fast Decision Guide

Question	Choose
Card ko surface se lift dikhana hai?	box-shadow
Text image ke upar readable banana hai?	text-shadow
Transparent logo/icon ke shape par shadow chahiye?	filter: drop-shadow()
Button pressed feel chahiye?	inset box-shadow
Keyboard focus visible banana hai?	:focus-visible + outline
Outline element se thoda door chahiye?	outline-offset
Realistic depth chahiye?	multiple shadows
Dark UI mein separation chahiye?	border + subtle shadow/glow

■ Best Practices - Yaad Rakhna

DO

Shadow opacity low rakho.
rgba/hsla colors use karo.
Design system shadow tokens banao.
Focus outline always visible rakho.
outline-offset se focus ring clean banao.
drop-shadow shape-based images ke liye use karo.

AVOID

outline: none without replacement.
Heavy black shadows everywhere.
Large animated shadows.
Paragraph text par strong text-shadow.
Random shadow values in every component.
Glow effects ka overuse.

> NEXT TOPICS TO COVER

Flexbox Deep Dive

CSS Grid

Positioning

Overflow

Z-index

Transitions

Animations

Forms Styling

